INCH-POUND
MIL-PRF-3098/84D
27 August 1997
SUPERSEDING
MIL-C-3098/84C
10 April 1995

PERFORMANCE SPECIFICATION SHEET

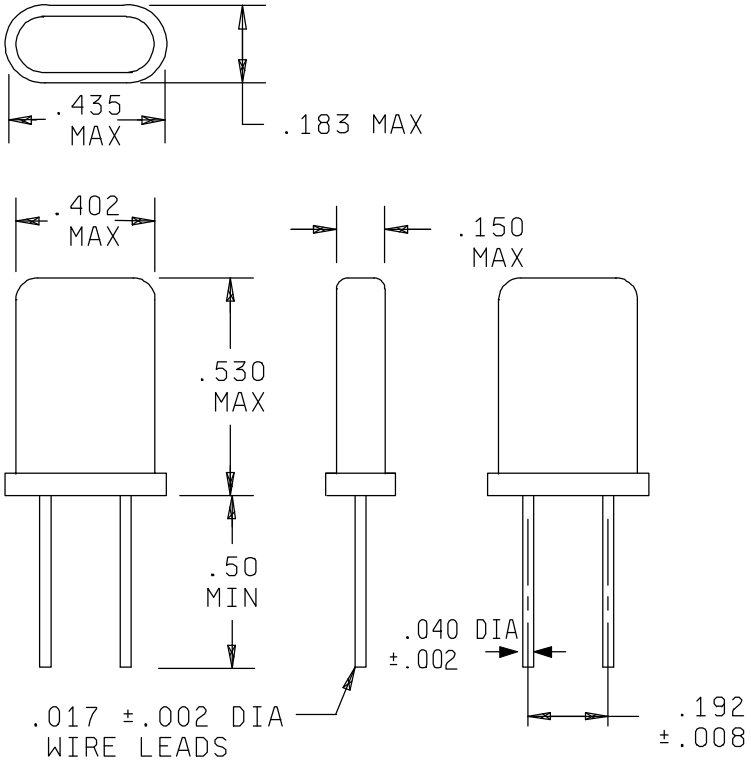
CRYSTAL UNIT, QUARTZ, CR108/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 50 MHz to 93 MHz; fifth mechanical overtone; controlled; series resonance.

Inches	mm
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.500	12.70
.530	13.46



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR108/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:44Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1. 1/

Frequency range: 50 MHz to 93 MHz, inclusive.

Capacitance, shunt: 5.5 pF, maximum.

Frequency stability: ±2.5 ppm.

Mode of oscillation: Fifth mechanical overtone.

Operating temperature range (controlled): +80°C to +90°C, inclusive.

Frequency tolerance: ±10 ppm (including frequency stability).

Equivalent resistance: 50 ohms, maximum.

Operable temperature range: -55°C to +80°C and +90°C to +105°C.

Reference temperature: +85°C ±1°C.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration high frequency: Method 204, MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±3 ppm.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

1/ These crystal units were previously fabricated using Air Force Technical Order 12-1-20 and/or Collins drawing numbers 289-2844 through 2858-00.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:44Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Air Force -85

Review activity:
Air Force - 99

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-0697-53)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:44Z
Check the source to verify that this is the current version before use.